


# APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

## B.Tech CSE 5th SEMESTER

**SUBMMITED TO:**

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# Lab Assignment 2: Largest, Smallest and Count

## Experiment No. 2.1 - Find the largest value

### Experiment Title:

Find the largest value

### Aim:

To find the largest value in an array with a loop.

### Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

### Theory:

To find the largest value, assume the first element is the largest and compare it with every other element, updating when a bigger value is found. The smallest value is found the same way with the comparison reversed. Counting uses a counter that is increased each time an element satisfies a condition.

### Algorithm / Procedure:

1. Set largest = marks[0].
2. Loop from index 1 to the end.
3. If marks[i] > largest, set largest = marks[i].
4. Print largest with the label Largest.

### Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 2, Task 1
// Question: Create the array int[] marks = {45, 67, 80, 52, 90}.
// Find the largest value with a loop and print it as "Largest:"
// followed by the value.

public class Main {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int largest = marks[0];

        for (int i = 1; i < marks.length; i++) {
            if (marks[i] > largest) {
                largest = marks[i];
            }
        }
```

```
        System.out.println("Largest: " + largest);
    }
}
```

## Input:

Fixed array declared in the program: marks = {45, 67, 80, 52, 90}.

## Output:

```
Largest: 90
```

**Time / Space Complexity:** Time O(n); Space O(1).

## Result:

The program was compiled and executed successfully. The largest value of the array was found correctly.

## Viva-Voce Questions:

- Why is largest initialised with marks[0]?
- Why does the loop start from index 1?
- What happens if all values are equal?

# Experiment No. 2.2 - Find the smallest value

## Experiment Title:

Find the smallest value

## Aim:

To find the smallest value in an array with a loop.

## Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

## Theory:

To find the largest value, assume the first element is the largest and compare it with every other element, updating when a bigger value is found. The smallest value is found the same way with the comparison reversed. Counting uses a counter that is increased each time an element satisfies a condition.

## Algorithm / Procedure:

1. Set smallest = marks[0].
2. Loop from index 1 to the end.
3. If marks[i] < smallest, set smallest = marks[i].
4. Print smallest with the label Smallest.

## Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 2, Task 2
// Question: Create the array int[] marks = {45, 67, 80, 52, 90}.
// Find the smallest value with a loop and print it as "Smallest:"
// followed by the value.

public class Main {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int smallest = marks[0];

        for (int i = 1; i < marks.length; i++) {
            if (marks[i] < smallest) {
                smallest = marks[i];
            }
        }

        System.out.println("Smallest: " + smallest);
    }
}
```

## Input:

Fixed array declared in the program: marks = {45, 67, 80, 52, 90}.

## Output:

```
Smallest: 45
```

**Time / Space Complexity:** Time O(n); Space O(1).

## Result:

The program was compiled and executed successfully. The smallest value of the array was found correctly.

## Viva-Voce Questions:

- How does this program differ from finding the largest value?
- What is the time complexity of finding the minimum?
- Can the smallest value be found without a loop?

# Experiment No. 2.3 - Count values greater than 60

## Experiment Title:

Count values greater than 60

## Aim:

To count the array values that are greater than 60.

## Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

## Theory:

To find the largest value, assume the first element is the largest and compare it with every other element, updating when a bigger value is found. The smallest value is found the same way with the comparison reversed. Counting uses a counter that is increased each time an element satisfies a condition.

## Algorithm / Procedure:

1. Set count = 0.
2. Loop over every element.
3. If marks[i] > 60, increase count by 1.
4. Print count with the label Count.

## Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 2, Task 3
// Question: Create the array int[] marks = {45, 67, 80, 52, 90}.
// Count the values greater than 60 and print the count as "Count:"
// followed by the number.

public class Main {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int count = 0;

        for (int i = 0; i < marks.length; i++) {
            if (marks[i] > 60) {
                count++;
            }
        }

        System.out.println("Count: " + count);
    }
}
```

## Input:

Fixed array declared in the program: marks = {45, 67, 80, 52, 90}.

## Output:

```
Count: 3
```

**Time / Space Complexity:** Time O(n); Space O(1).

## Result:

The program was compiled and executed successfully. The number of values greater than 60 (3) was counted correctly.

## Viva-Voce Questions:

- What does count++ do?
- Which values in the array are greater than 60?
- How would you count values less than 60?